

Title

Temporal restriction of feeding windows as a dietary intervention in metabolic syndrome: rationale and design of a randomized controlled feeding trial

Abstract

Rationale: Structured dietary interventions addressing meal timing may influence appetite signaling and cardiometabolic risk factors. This study compares the effects of time-restricted versus conventional eating patterns on metabolic and behavioral measures.

Methods: A 12-week randomized feeding trial will enroll 40 overweight/obese adults aged 50-70 with prediabetes or treated type 2 diabetes. One group will follow a 10-hour eating window; the control group will eat during a standard 16-hour window. Both groups receive nutritionally equivalent calories. Assessments will include three-factor eating behavior questionnaire responses, fasting hormonal markers (leptin, adiponectin), inflammatory markers (C-reactive protein), and cortisol. Oral glucose tolerance testing and ghrelin measurement will be performed. Statistical comparisons will use non-parametric tests for within- and between-group differences. The trial will examine whether meal timing, independent of calorie reduction, influences appetite physiology and eating patterns.